Md Mahmudur Rahman

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Education

- **Ph.D. in Information Systems** Aug 2019 - Aug 2024
  University of Maryland, Baltimore County (UMBC), USA
- **Master's in Information Systems** Aug 2019 - May 2023
  University of Maryland, Baltimore County (UMBC), USA CGPA: 3.87/4.00
- **Master's in Statistics** Jan 2016 – Mar 2017
  University of Dhaka, Bangladesh GPA: 3.89/4.00
- **Bachelor's in Statistics, Biostatistics and Informatics** Jan 2011 – Nov 2015
  University of Dhaka, Bangladesh CGPA: 3.78/4.00

Work Experience

- **Applied Scientist**, Central Seller Fulfillment Team, Amazon, USA Sep 2024 – Present
  - Generated **$46.7M annual GMS lift (165x ROI)** by designing and deploying a Neural Causal Model that replaced a legacy DML pipeline, expanding seller recommendation coverage from 55% to 95% across 10 global marketplaces; published at WSDM CausalBench'26
  - Achieved the **only cold-start WAPE below 1.0** (0.66 vs. 1.06 next-best baseline) with **82.8% median error reduction** and **19.1% MAE improvement**, by engineering a cluster-based transfer approach over 350M+ training samples and 1.6B+ inference records—while preserving best-in-class warm-start performance (WAPE: 0.67)
  - Accelerated cross-org adoption by leading weekly syncs with 3+ teams and aligning on evaluation standards, driving **two teams to adopt** the model within 2 months of launch
  - Launched explainability framework across multiple marketplaces, engaging **15K+ sellers** by developing a browse-node and price-based similarity engine that delivers personalized, data-backed recommendation insights
  - Improved SmartBiz theme recommendation accuracy (**Hit@1: 30.1% → 40.7%, MRR@3: 0.44 → 0.56**) by designing two automated evaluation frameworks—an LLM-powered screenshot analysis system and a 2,000-conversation simulation pipeline—reducing evaluation cycles from weeks to hours
  - Architected production ML infrastructure on **AWS Step Functions, EMR, and SageMaker**, processing large-scale datasets for weekly batch inference and training with high availability and fault tolerance
- **Applied Scientist Intern**, PXTCS team, Amazon, USA Feb 2024 – May 2024
  - Developed expertise in **QLoRA** Large Language Model finetuning, working with state-of-the-art models like **Meta-Llama-3-8B** and **Mistral-instruct-V2.0**
  - Implemented **custom loss functions** and advanced **prompt engineering** techniques, achieving measurable performance improvements in model accuracy and inference quality
  - Built high-performance **data preprocessing pipelines** processing large-scale datasets with optimized **GPU utilization**, reducing training time and computational costs
  - Owned development of **deep survival models** for OPS attrition forecasting, delivering **5% performance improvement** over existing system and driving data-driven retention strategies
- **Graduate Research Assistant**, UMBC, USA Aug 2019 – Dec 2023
  - Pioneered novel **fairness definitions** and algorithmic techniques for **time-to-event data** analysis, establishing new standards for equitable healthcare predictions
  - Invented **privacy-preserving federated learning** frameworks for distributed **survival analysis**, enabling secure multi-institutional collaboration while maintaining data privacy
  - Delivered **interpretable machine learning** models for survival analysis, providing clinicians with actionable insights for patient care decisions
  - Developed breakthrough techniques for complex survival problems including **competing risk analysis** and **multi-state modeling**, resulting in **14 publications** in top-tier AI venues (**AAAI, KDD, SDM, CIKM**)

- **Assistant Director**, The Central Bank of Bangladesh — Apr 2018 - Aug 2019
  - Delivered **data-driven insights** on Bangladesh's export performance by analyzing national **Export Receipts & Central Account** data using **Enterprise Data Warehouse**, directly informing **monetary policy** decisions
  - Owned **quarterly and annual export performance reviews** for Goods & Services, providing critical economic intelligence to **senior leadership** and government stakeholders
  - Built and led the **first Data Science section** at Bangladesh Bank, establishing **data-driven culture** and analytical capabilities that transformed departmental decision-making processes

Publications

- **Fair Federated Survival Analysis**
  Md Mahmudur Rahman and Sanjay Purushotham
  AAAI - AAAI Conference on Artificial Intelligence 2025 [**Paper**]
- **Federated Competing Risk Analysis**
  Md Mahmudur Rahman and Sanjay Purushotham
  CIKM - Conference on Information and Knowledge Management 2023 [**Paper**]
- **FedPseudo: Privacy-Preserving Pseudo Value-Based Deep Learning Models for Federated Survival Analysis**
  Md Mahmudur Rahman and Sanjay Purushotham
  KDD - ACM Conference on Knowledge Discovery and Data Mining 2023 [**Paper**] [**Code**]
- **Communication-Efficient Pseudo Value-Based Random Forests for Federated Survival Analysis**
  Md Mahmudur Rahman and Sanjay Purushotham
  AAAI - AAAI Fall Symposium Series - SPACA: Survival Prediction Algorithms, Challenges & Applications 2023 [**Paper**]
- **Multi-state Survival Analysis using Pseudo value-based Deep Neural Networks**
  Md Mahmudur Rahman and Sanjay Purushotham
  SDM - SIAM International Conference on Data Mining 2023 [**Paper**]
- **Fair and interpretable models for survival analysis**
  Md Mahmudur Rahman and Sanjay Purushotham
  KDD - ACM Conference on Knowledge Discovery and Data Mining 2022 [**Paper**] [**Code**]
- **DeepPseudo: pseudo value based deep learning models for competing risk analysis**
  Md Mahmudur Rahman, Koji Matsuo, Shinya Matsuzaki, and Sanjay Purushotham
  AAAI - AAAI Conference on Artificial Intelligence 2021 [**Paper**] [**Code**]
- **PseudoNAM: A Pseudo Value Based Interpretable Neural Additive Model for Survival Analysis**
  Md Mahmudur Rahman and Sanjay Purushotham
  AAAI - AAAI 2021 Fall Symposium on Human Partnership with Medical AI: Design, Operationalization, and Ethics (AAAI-HUMAN 2021) [**Paper**]

Abstract and Workshop Presentations

- **Federated learning for competing risk analysis in healthcare**
  Md Mahmudur Rahman and Sanjay Purushotham
  KDD - International Workshop on Federated Learning for Distributed Data Mining 2023 [**Paper**] [**Link**]
- **Federated Survival Analysis with Competing Events (Abstract)**
  Md Mahmudur Rahman and Sanjay Purushotham
  CISS - 57th Annual Conference on Information Sciences and Systems 2023 [**Link**]
- **FedPseudo: Pseudo value-based Deep Learning Models for Federated Survival Analysis**
  Md Mahmudur Rahman and Sanjay Purushotham
  KDD - DSHealth: Workshop on Applied Data Science for Healthcare 2022 [**Paper**] [**Link**]

- **Pseudo value-based Deep Neural Networks for Multistate Survival Analysis**
  Md Mahmudur Rahman and Sanjay Purushotham
  KDD - DSHealth: Workshop on Applied Data Science for Healthcare 2022 [**Paper**] [**Link**]
- **A Pseudo Value Based Interpretable Neural Additive Model for Survival Analysis**
  Md Mahmudur Rahman and Sanjay Purushotham
  AAAI - Workshop on Trustworthy AI for Healthcare 2022 [**Link**]
- **DeepPseudo: A Deep learning approach based on Pseudo values for Competing Risk Analysis**
  Md Mahmudur Rahman and Sanjay Purushotham
  KDD - DSHealth: Workshop on Applied Data Science for Healthcare 2022 [**Link**]

Skills

- **Programming Languages:** Python, R, SQL, Bash
- **ML/AI Frameworks:** PyTorch, HuggingFace Transformers, TensorFlow, Keras, scikit-learn, NumPy, Pandas
- **AWS Cloud Services:** SageMaker, EMR, Step Functions, Lambda, S3, Bedrock, Amazon Q
- **Big Data & MLOps:** Apache Spark, PySpark, Cradle, Docker, Git, Linux, Jupyter, MLflow
- **Statistical Tools:** STATA, SPSS, SAS, R Studio
- **Research Expertise:** Large Language Models, Causal Inference, Survival Analysis, Federated Learning, Algorithmic Fairness, Explainable AI, Agentic AI Architectures and Evaluation

Teaching and Mentoring Experience

- **Graduate Teaching Assistant** — Jan 2023 – May 2023
  Department of Information Systems, University of Maryland-Baltimore County
  - Supported **IS 757 Deep Learning** course under Dr. James Foulds, grading assignments and exams for **40 students** and conducting targeted discussion sessions to reinforce complex course concepts
- **Mentor**, NSF-REU Project — Jun 2021 – Aug 2021
  University of Maryland-Baltimore County
  - Mentored an undergraduate researcher through an **NSF-REU** summer project on **medical image segmentation**, guiding experiment design and model development to successful project completion
- **Lecturer**, Bangladesh University of Business and Technology — Jun 2017 – Sep 2017
  - Designed and delivered **three statistics courses** covering **business statistics, regression analysis, probability, and time series analysis**
  - Monitored student progress through regular assessments and facilitated group discussions to develop analytical reasoning and communication skills
- **Graduate Teaching Assistant** — Jan 2017 – Apr 2017
  East West University, Dhaka, Bangladesh
  - Taught **R programming** for **Design of Experiments** and **Multivariate Analysis** to **20 graduate students**, designing course materials and grading all assignments and exams
- **Instructor** — Sep 2016 – Dec 2016
  East West University, Dhaka, Bangladesh
  - Taught **statistical learning with SPSS and R** to **120 students** across three sections, managing end-to-end course delivery including assessments and student support

Professional Activities and Recognition

- **ACM SIGKDD 2023 Student Travel Award**
- **SIAM Student Travel Award**, SIAM International Conference on Data Mining, 2023
- **Peer-reviewer** in Biostatistics journal and AISTATS, AIMHC and CISS conferences
- **1st Place - PhD/Postdoc Completed Research Track**, IS Research Symposium, UMBC, 2022
- **Student Scholarship**, ACM SIGKDD-2022
- **Volunteered** at AAAI-2021 and ACM SIGKDD-2022 conferences
- **Student Scholarship**, AAAI-2021 and AAAI-2022
- **Best Poster Award**, IS Poster Day, UMBC, 2021
- **Dean's Award**, University of Dhaka, Bangladesh, 2016
- **NST Fellowship for Research**, Ministry of Science and Technology, Bangladesh, 2016